

Solving Word Problems Involving Unit Conversions

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Unit conversion is used to express the same measurement using a different unit.

Identify the Given and Required

Read the word problem carefully and identify the **given**, the **required**, and the **unit** needed in the final answer.

Use a Conversion Factor

Use the appropriate **conversion factor** to change one unit into another.

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ L} = 1000 \text{ mL}$$

Cancel Unwanted Units

Make sure unwanted units **cancel out** during the conversion process.

$$2.8 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} = 2800 \text{ m}$$

Write the Correct Unit

Always write the **correct unit** with your final answer.

$$2.8 \text{ km} = 2800 \text{ m}$$

The final answer is 2800 meters.

Examples

Example 1

Maria walked 2.8 km from her house to school.
How many meters did she walk?

$$2.8 \text{ km} \times 1000 = 2800 \text{ m}$$

Therefore, Maria walked 2800 meters.

Example 2

A bottle contains 3.5 L of juice. Express the amount in milliliters.

$$3.5 \text{ L} \times 1000 = 3500 \text{ mL}$$

Therefore, the bottle contains 3500 mL.

Example 3

A carpenter needs a wooden plank that is 250 cm long. Express its length in meters.

$$250 \text{ cm} \div 100 = 2.5 \text{ m}$$

Therefore, the plank is 2.5 meters long.

Practice Exercises

Practice 1

Maria walked 2.8 km from her house to school.
How many meters did she walk?

Practice 2

A bottle contains 3.5 L of juice. Express the amount in milliliters.

Practice 3

A carpenter needs a wooden plank that is 250 cm long. Express its length in meters.

Practice 4

A motorcycle traveled 18 000 m. Express the distance in kilometers.

Practice 5

A baby weighs 4.2 kg. Express the weight in grams.

Practice 6

A water tank holds 7 500 mL of water. Express the capacity in liters.

Practice 7

A recipe requires 1.5 kg of flour. If flour costs ₱80 per kilogram, how much will the flour cost?

Practice 8

A runner completed a race of 5 000 m. How many kilometers did the runner complete?

Activities

Activity 1

Carlo biked 4.6 km around the park. How many meters did he travel?

Activity 2

A container holds 2.8 L of milk. Express the amount in milliliters.

Activity 3

A ribbon measures 360 cm. Express its length in meters.

Activity 4

A train traveled 42 000 m. Express the distance in kilometers.

Activity 5

A sack of rice weighs 25 kg. Express the weight in grams.

Activity 6

A pitcher contains 9 000 mL of water. Express the capacity in liters.

Activity 7

A fruit vendor bought 2.5 kg of apples. If apples cost ₱120 per kilogram, how much did the vendor pay?

Activity 8

A delivery truck traveled 86 km. Express the distance in meters, then determine how many meters the truck traveled.

Activity 9

A container holds 3.6 L of fruit juice. It was poured equally into 8 bottles. How many milliliters of juice were placed in each bottle?

Activity 10

A cyclist completed a route measuring 12 000 m. How many kilometers did the cyclist travel?

Thank you!